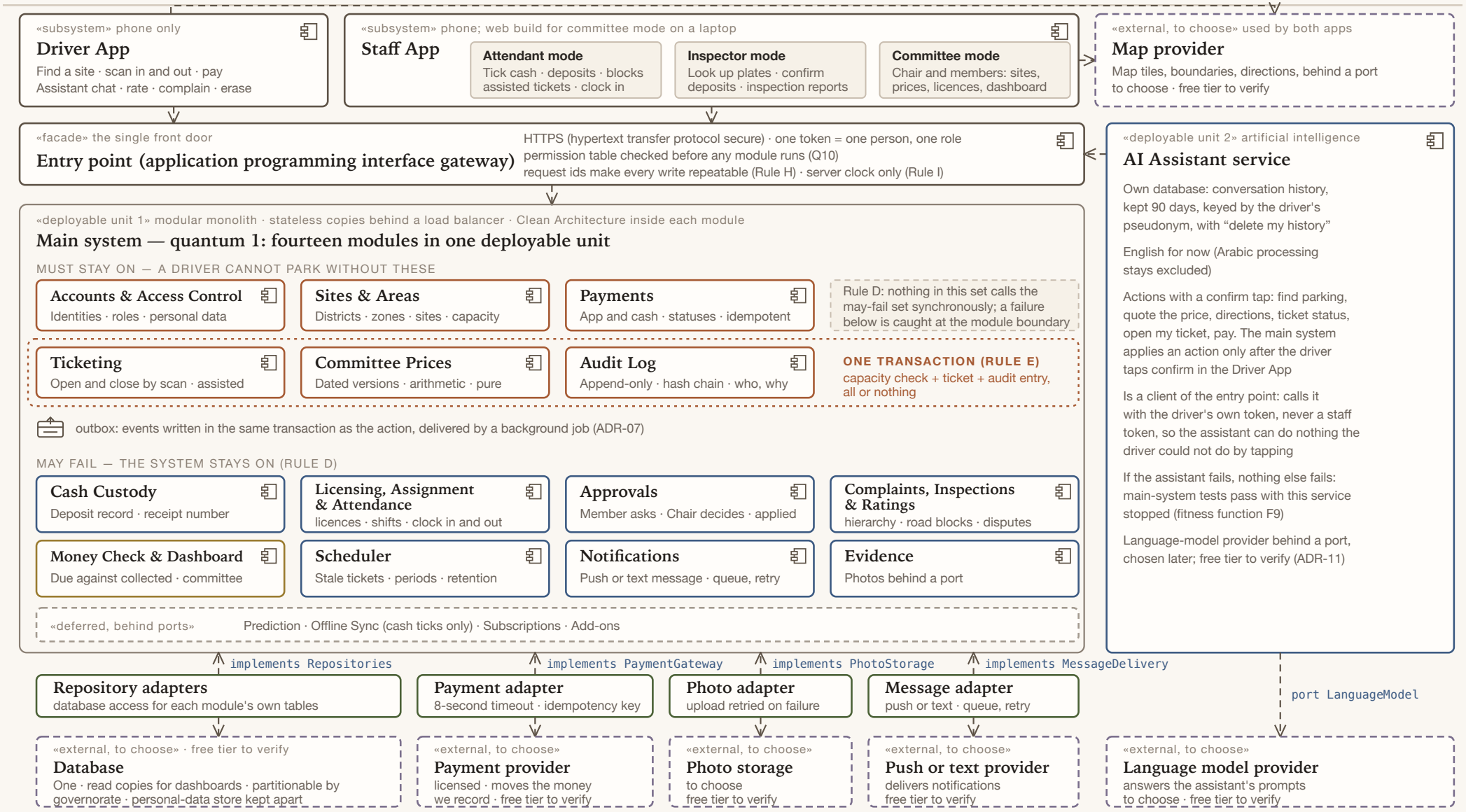
Street Parking Management System — Component Architecture

UML (Unified Modeling Language) component diagram · level 1: context, two deployable units and ports · architecture V3 · 4 November 2026



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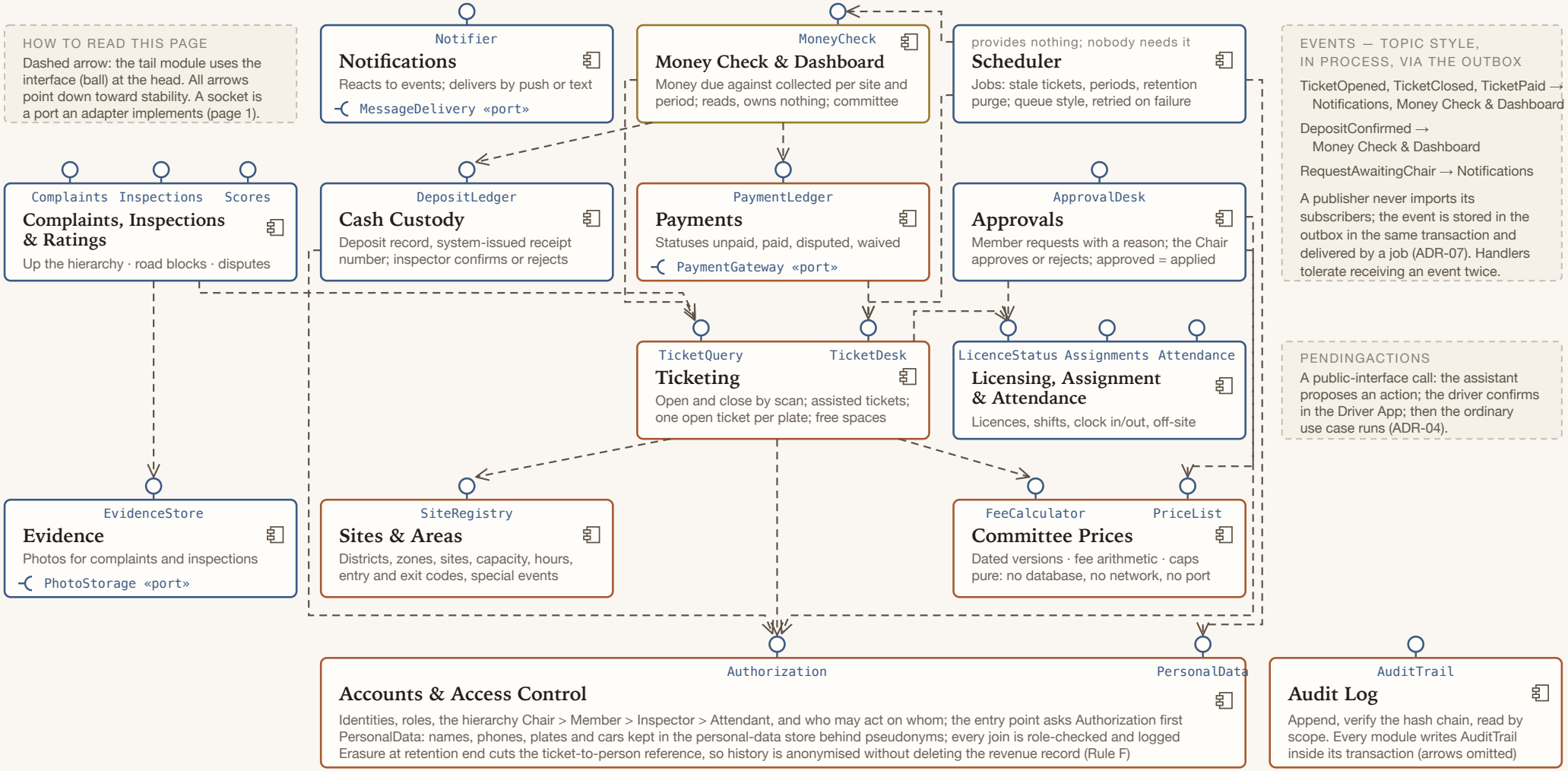
Must stay on · May fail, the system stays on · Committee only · Background and adapters · Deferred, behind ports · External, to choose · UML component

Dependency: the tail needs the head · One transaction (Rule E) · Outbox: the event is stored with the write and delivered by a job (ADR-07)

Architectural point: the main system is one deployable unit because integrity (rank 1) and accountability (rank 2) live in one database transaction — the capacity check, the ticket and its audit entry succeed or fail together, which no network call can promise. The assistant is a separate unit because its failure must touch nothing: the assistant may be slow, crash or be redeployed while tickets keep opening. Scale comes from running more stateless copies of the same unit behind the load balancer, never from splitting the unit; the only lock is one site's row, held for milliseconds (ADR-03).

Modules and Their Interfaces

UML component diagram · level 2: inside deployable unit 1 · provided interfaces (balls), required ports (sockets) and the dependencies that use them · 4 November 2026



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- Provided interface (ball), named inside the box under the ball
- Dependency: the tail module uses the interface at the head
- Required port: the module owns it, an adapter outside implements it (page 1)
- UML component
- Must stay on: a driver cannot park without it (six modules)
- May fail, the system stays on (Rule D; eight modules)
- Committee only (also may fail)

Architectural point: Committee Prices has no required interface, so it holds no database, network or port code, and every fee rule is tested with plain numbers in milliseconds (fitness function F1).
The must-stay-on set (brown) never imports the may-fail set (blue): a failure in Notifications, Approvals or the dashboard is caught at the module boundary and a ticket still opens (Rule D, fitness function F8);
one arrow, Ticketing → LicenceStatus, crosses that line as written in the interface catalogue (Q12) and is flagged for Zeyad's decision. No cycle exists: every arrow points downward or sideways, the Chair's
approval reaches Licensing and Prices as a call from Approvals above them, and events travel upward only through the outbox, where a publisher never imports its subscribers (Acyclic Dependencies Principle).